Primitive Roots

Patrick Was

1 Primitive Roots

1.1 Definition

Definition 1.1.1 — An integer a is said to be a *primitive root* mod n if for every number coprime to n, there is an integer b such that $a^b \equiv k \mod n$
Note: By standard of mod, $k < n$.

However, when talking about primitive roots, it is more useful in general to consider a group theory-related definition. This is helpful when proving certain theorems as well as helping recognize patterns within primitive roots.

Definition 1.1.2 — Define

$$\mathbb{Z}_n^\times = \{a \in \mathbb{N} | 1 \leq a < n, \gcd(a, n) = 1\}$$

Then, a *primitive root* mod n is an element $g \in \mathbb{Z}_n^\times$ such that every element $b \in \mathbb{Z}_n^\times$ can be written as g^k mod n, for some $k \in \mathbb{Z}$

With this definition, we could say that a primitive root is a group generator of $\mathbb{Z}_n^\times$.

1.2 Examples

On the left side is an example of a primitive root mod 5, and on the right is an example of an integer that is not a primitive root mod 5.

$$\begin{aligned} 3^1 &\equiv 3 \mod 5 \\ 3^2 &\equiv 4 \mod 5 \\ 3^3 &\equiv 2 \mod 5 \\ 3^4 &\equiv 1 \mod 5 \end{aligned} \qquad \begin{aligned} 4^1 &\equiv 4 \mod 5 \\ 4^2 &\equiv 1 \mod 5 \\ 4^3 &\equiv 4 \mod 5 \\ 4^4 &\equiv 1 \mod 5 \end{aligned}$$

Because the set of integers less than 5 relatively prime to it is $\{1, 2, 3, 4\}$, each of these numbers must appear as a k in the definition. This is true for 3:

$$\begin{aligned} 3^4 &\equiv 1 \mod 5 \\ 3^3 &\equiv 2 \mod 5 \\ 3^1 &\equiv 3 \mod 5 \\ 3^2 &\equiv 4 \mod 5 \end{aligned}$$

But we see that for 4, we only achieve 1 and 4. This can be explained using fundamental number theory: knowing that $4 \equiv -1 \mod 5$, we know that $4^k \equiv (-1)^k \mod 5$, so the value will be alternating between $1 \mod 5$ and $-1 \mod 5$, or $4 \mod 5$.

With this logic, we know that $n-1$ will never be a primitive root mod n.

2 Chapter ϕve

2.1 Computing Euler's Totient Function

> **Definition 2.1.1** — Let *Euler's totient function* $\phi : \mathbb{N} \to \mathbb{N}$ (sometimes written as φ) be the number of positive integers less than or equal to n that are relatively prime to n.

The function is clearly defined for prime inputs: if p is a prime,

$$\phi(p) = p - 1$$

because for all $1 \leq q < p$, $\gcd(p, q) = 1$.

A general formula for $\phi(p^k)$ with prime p is also very approachable. It is significantly easier to do complementary counting here; the only $1 \leq q < p$ for which $\gcd(p, q) \neq 1$ are multiples of p. There are $\frac{p^k}{p} = p^{k-1}$ of these multiples, so

$$\phi(p^k) = p^k - p^{k-1}$$

Though it's nice to have formulas for prime numbers, the function is defined for all natural numbers, so it would be nice to have a general formula for any natural n.

A relatively simple observation to make occurs when we take the product of two primes. If we let this number be ab, with a and b being the two primes multiplied together, we can develop a pattern using complementary counting:

$$\begin{aligned}\phi(ab) &= ab - a - b + 1 \\ &= (a-1)(b-1) \\ &= \phi(a)\phi(b)\end{aligned}$$

We simply subtract the number of multiples of b (which is a), the number of multiples of a (which is b), and since we overcounted by 1 (we would have counted ab twice), we add 1. This factors to $(a-1)(b-1)$, which is equivalent to $\phi(a)\phi(b)$.

However, we can generalize this to any coprime a and b. We can generalize our previous result because the same logic will apply; we exclude the multiples of a, multiples of b, and fix any overlap (of which there will only be 1, since they are coprime), which just gives us $\phi(a)\phi(b)$

> **Definition 2.1.2** — Let f be a function such that $f(ab) = f(a)f(b)$ for positive a, b, where $\gcd(a, b) = 1$. Then, f is called a *multiplicative function.*
> **Note:** This is the standard definition in the branch of number theory. Generally, multiplicative functions do not have the restriction of two coprime factors, but here we will call multiplicative functions without this restriction *completely multiplicative functions.*

It's pretty clear that with this generalization, we can compute $\phi(n)$ for any $n \in \mathbb{N}$—though the process might be daunting for some values, as we will get to later. However, we can further generalize our results from the previous conclusion to compute $\phi(n)$ without even using ϕ.

If we notice that we can write $\phi(n)$ as $\phi(p_1^{a_1} p_2^{a_2} p_3^{a_3} \ldots)$, where $p_k^{a_k}$ is a prime factor of n, we can come up with an interesting computation.

Theorem 2.1.1

$$\phi(n) = n\left(1 - \frac{1}{p_1}\right)\left(1 - \frac{1}{p_2}\right)\left(1 - \frac{1}{p_3}\right)\ldots\left(1 - \frac{1}{p_k}\right)$$

or equivalently

$$\phi(n) = n\prod_{p|n}\left(1 - \frac{1}{p}\right)$$

where p_k is a prime factor of n.

Proof.

$$\begin{aligned}
\phi(n) &= \phi(p_1^{a_1} p_2^{a_2} p_3^{a_3} \ldots p_k^{a^k}) \\
\phi(p_1^{a_1} p_2^{a_2} p_3^{a_3} \ldots p_k^{a^k}) &= \phi(p_1^{a_1})\phi(p_2^{a_2})\phi(p_3^{a_3})\ldots\phi(p_k^{a_k}) \\
\phi(p_1^{a_1})\phi(p_2^{a_2})\ldots\phi(p_k^{a_k}) &= (p_1^{a_1} - p_1^{a_1-1})(p_2^{a_2} - p_2^{a_2-1})\ldots(p_k^{a_k} - p_k^{a_k-1}) \\
&(p_1^{a_1-1}(p_1 - 1))(p_2^{a_2-2}(p_2 - 1))\ldots(p_k^{a_k-1}(p_k - 1)) \\
&p_1^{a_1}\left(1 - \frac{1}{p_1}\right)p_2^{a_2}\left(1 - \frac{1}{p_2}\right)\ldots p_k^{a_k}\left(1 - \frac{1}{p_k}\right)
\end{aligned}$$

Finally, because $p_1^{a_1} p_2^{a_2} \ldots p_k^{a_k} = n$:

$$\phi(n) = n\left(1 - \frac{1}{p_1}\right)\left(1 - \frac{1}{p_2}\right)\left(1 - \frac{1}{p_3}\right)\ldots\left(1 - \frac{1}{p_k}\right)$$

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Corrolary. $\phi(n)$ for any $n > 2$ is always even.

Proof.

If n is a power of 2 greater than 1 $\implies \phi(2^k) = 2^k - 2^{k-1}$, which is always even for $k > 1$.

If n is composed of an odd prime p, we can rewrite this factor in the formula as

$$1 - \frac{1}{p} = \frac{p-1}{p}$$

$p - 1$ will always be even since p is prime, and since we are multiplying n by an even power, $\phi(n)$ must be even.

■

Though the computation has been greatly reduced, there is more we can do. Say that we know $\phi(8)$ and $\phi(10)$ and desire to compute $\phi(80)$. It wouldn't be too difficult to write $\phi(80)$ as a product of coprime factors; as a matter of fact, $\phi(16)$ and $\phi(5)$ would work just fine. However, as n increases, it will become more challenging to find coprime factors of n, especially if finding the prime factorization is too daunting.

Theorem 2.1.2

$$\phi(ab) = \phi(a)\phi(b)\frac{\gcd(a,b)}{\phi(\gcd(a,b))}$$

Proof.

$$\begin{aligned}
\phi(ab) &= ab\prod_{p|ab}\left(1-\frac{1}{p}\right)\\
&= ab\frac{\prod_{p|a}\left(1-\frac{1}{p}\right)\prod_{p|b}\left(1-\frac{1}{p}\right)}{\prod_{p|\gcd(a,b)}\left(1-\frac{1}{p}\right)}\frac{\gcd(a,b)}{\gcd(a,b)}\\
&= \phi(a)\phi(b)\frac{\gcd(a,b)}{\phi(\gcd(a,b))}
\end{aligned}$$

We use $\gcd(a,b)$ because there may be duplicate prime factors accounted for in the numerator, since a and b are not necessarily coprime.

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2.2 Cardinality of the Set

The connection between $\mathbb{Z}_n^\times$ and $\phi(n)$ is pretty clear: $|\mathbb{Z}_n^\times| = \phi(n)$, since $\phi(n)$ simply counts the number of elements in the particular set.

Theorem 2.2.1

Considering our previous definition of $\mathbb{Z}_n^\times$:

$$|\mathbb{Z}_n^\times| = \phi(n)$$

3 More on Primitive Roots

3.1 Existence

Theorem 3.1

There are primitive roots mod n iff $n = 1, 2, 4, p^k$ or $2p^k$, where p is an odd prime and $k \in \mathbb{N}$.

In order to prove this theorem, it's most helpful to first address when there are no primitive roots and then prove for the remaining values of n that there do exist primitive roots.

Lemma 1. Let $m, n \in \mathbb{N}$. If $\gcd(m,n) = 1$ and $m, n \geq 3$, then there are no primitive roots mod mn.

Proof.
Consider some $a \in \mathbb{N}$ such that $\gcd(a, mn) = 1 \implies \gcd(a,m) = \gcd(a,n) = 1$.
Both $\phi(m)$ and $\phi(n)$ must be even. By Euler's Theorem:

$$
\begin{aligned}
a^{\frac{\phi(m)\phi(n)}{2}} &= (a^{\phi(m)})^{\frac{\phi(n)}{2}} \equiv 1 \mod m \\
&= (a^{\phi(n)})^{\frac{\phi(m)}{2}} \equiv 1 \mod n
\end{aligned}
$$

By the Chinese Remainder Theorem:

$$a^{\frac{\phi(n)\phi(m)}{2}} \equiv 1 \mod mn$$

But,

$$\frac{\phi(m)\phi(n)}{2} = \frac{\phi(mn)}{2} < \phi(mn)$$

Because the order of a is less than $\phi(mn)$, a cannot be a primitive root mod mn.

□

This helps us see that n does not have a primitive root if it is divisible by two odd primes or can be expressed as $2^k p^l$, where $k \geq 2$ and p is an odd prime.

Lemma 2. For any $k \in \mathbb{N}$, there is a primitive root mod $2p^k$ where p is an odd prime.

Proof.

$$\phi(2p^k) = \phi(p^k)$$

If g is odd:

$$
\begin{gathered}
\gcd(g, 2p^k) = 1 \\
\operatorname{ord}_{2p^k}(g) | \phi(p^k) \quad \text{and} \quad g^{\operatorname{ord}_{2p^k}(g)} \equiv 1 \mod p^k \\
\implies \operatorname{ord}_{2p^k}(g) = \phi(p^k) = \phi(2p^k)
\end{gathered}
$$

So, g is a primitive root mod $2p^k$. If g is even, $g + p^k$ is odd:

$$g + p^k \equiv g \mod p^k$$

$g + p^k$ is an odd primitive root and the previous steps follow.

□

With these two lemmas, the theorem follows.

3.2 Order

Unfortunately, there is no direct way of computing primitive roots. It is generally reduced to trial and error, though there are methods that shorten this computation.

> **Definition 3.1.1** — Define $\operatorname{ord}_n(g)$ to be the smallest $k > 0$ such that $g^k \equiv 1 \mod n$.

An alternative definition to the one just constructed relates to *periods* of powers modulo n. Because the power k is strictly greater than 0, we can also define $\operatorname{ord}_n(g)$ as the period of g, which simply means how many powers must be cycled through before finding an equivalent modulus.

By **Euler's Theorem** that $g^{\phi(n)} \equiv 1 \mod n$, so by definition 3.1.1, $k \le \phi(n)$, so $\phi(n)$ is the upper bound of k.

We know that $\text{ord}_n(g) | \phi(n)$. This is pretty clear to see, because $\text{ord}_n(g) \le \phi(n)$ but it must be true that for some $y \in \mathbb{N}$, $y\text{ord}_n(g) = \phi(n)$, which ties back to the alternative definition.

thm

g is a primitive root mod n iff $\text{ord}_n(g) = \phi(n)$.

4 Other Useful Theorems

Theorem 4.1

$$\text{ord}_n(a^k) = \frac{\text{ord}_n(a)}{\gcd(k, \text{ord}_n(a))}$$

Proof.

Let $d = \gcd(k, \text{ord}_n(a))$

$$\begin{aligned}
(a^k)^{\frac{\text{ord}_n(a)}{d}} &= \\
&= (a^{\text{ord}_n(a)})^{\frac{k}{d}} \\
&= 1^{\frac{k}{d}} \\
&= 1 \\
&\equiv 1 \mod n
\end{aligned}$$

$\implies \text{ord}_n(a^k) \mid \frac{\text{ord}_n(a)}{d}$

$$a^{k\text{ord}_n(a^k)} \equiv 1 \mod n$$

$\implies \text{ord}_n(a) \mid k\text{ord}_n(a^k)$
$\implies \frac{\text{ord}_n(a)}{d} \mid \frac{k}{d}\text{ord}_n(a^k)$

Because $\frac{\text{ord}_n(a)}{d} \nmid \frac{k}{d}$, it must divide $\text{ord}_n(a^k)$.

Since we have divisibility both ways, $\frac{\text{ord}_n(a)}{d} = \text{ord}_n(a^k)$

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Theorem 4.2

If there are primitive roots mod n, then there are exactly $\phi(\phi(n))$ primitive roots mod n.

This theorem is a corrolary of the previous one, but influential enough to be its own theorem.

Proof.

Let g be a primitive root mod n.

Then, any other primitive root is in the set $\{1, r, r^2, \ldots, r^{\phi(n)-1}\} \implies r^k, 0 < k < \phi(n)$

$$\text{ord}_n(r^k) = \frac{\text{ord}_n(r)}{\gcd(k, \text{ord}_n(r)} = \frac{\phi(n)}{\gcd(k, \phi(n))} = \phi(n)$$

$\implies \gcd(k, \phi(n) = 1 \to$ There are $\phi(\phi(n))$ such numbers

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4.1 Examples

As an example, we can find all primitive roots mod 18. Because $18 = 2(3^2)$, we know that there must exist primitive roots mod 18. If we look at

$$a^b \equiv k \mod n$$

we know we are looking for a valid a, or a valid primitive root. Because our conditions are that $\text{ord}_n(a)|\phi(n) = 6$ and $\gcd(a, n) = 1$, we can test some possible values, denoted using hats:

$$\hat{k} \in \{1, 2, 3, 6\}$$

$$\hat{a} \in \{1, 5, 7, 11, 13, 17\} = \mathbb{Z}_{18}^{\times}$$

We can discount 1 because its order will always be 1. If we let each element of set S correspond to an element of $\mathbb{Z}_{18}^{\times}$, we get:

$$S = \{1, 6, 3, 6, 3, 2\}$$

Unsurprisingly, 17 doesn't work, as we've shown before. The primitive roots mod 18 are 5 and 11.

Using Theorem 4.2, we can see that there are exactly $\phi(\phi(8)) = 2$ primitive roots mod 8, which is exactly what we want. This is helpful because we know we could stop at 11 instead of computing all the way to 17.

As another example, we can attempt to find the primitive roots mod 8. We can immediately conclude that there are NO primitive roots using the existence theorem, but we can make an attempt to see what happens if we try.

$$\phi(8) = 4$$

$$\hat{k} \in \{1, 2, 4\}$$

$$\hat{a} \in \{1, 3, 5, 7\} = \mathbb{Z}_8^{\times}$$

We see that $3^2, 5^2$, and 7^2 are all 1 mod 8, which means that their orders are all 2. Because $2 \neq 4$, there are no primitive roots mod 8.

5 Discrete Logarithms

> **Definition 5.1** — If $a \equiv b^d \mod m$, then d is the *discrete logarithm* of a modulo m to the base b,
>
> $$d = \log_b(a)$$

Properties of logarithms transcribe to discrete logarithms: if k is the order of b mod m,

$$\log_b(ac) \equiv \log_b(a) + \log_b(c) \mod k$$

$$\log_b(a^r) \equiv r\log_b(a) \mod k$$

This is generally very useful in **RSA encryption** (because, as we've seen before, there is no closed form for all primitive roots, especially as numbers get more complicated) and occasionally in solving some equivalencies.

$$x^4 \equiv 9 \mod 11$$
$$\log_2(x^4) \equiv \log_2(9) \mod 10$$
$$2\log_2(x) \equiv 3 \mod 5$$
$$\log_2(x) \equiv 4 \mod 5 \implies 4, 9 \mod 10$$
$$x \equiv 2^4, 2^9 \mod 11 \to x \equiv 5, 6 \mod 11$$

6 Primitive Roots of Unity

Definition 6.1 — ζ is a primitive n^{th} root of unity iff

$$\zeta^n = 1, \zeta^k \neq 1, 0 < k < n$$

From this definition, we know that 1 will only be a primitive root for $n = 1$, though generally, the first primitive root of unity is not particularly helpful.

The primitive 4^{th} roots of unity are $\pm i$, because there is no $0 < k < n$ such that $i^k = 1$ except for $k = 4 = n$. 1 and -1 work for $k = 1$ and $k = 2$ respectively, and are thus not primitive roots.

Theorem 6.1

The primitive n^{th} roots of unity are of the form

$$\zeta_n = e^{\frac{2\pi i k}{n}}, k \in \mathbb{Z}_n^\times$$

and there are $\phi(n)$ primitive n^{th} roots of unity.

Proof.

ζ_n^k is only primitive if $k \mid n$ by the same order arguments as developed earlier.

$\gcd(k, n) = 1$ so that $(\zeta_n^k)^{\frac{n}{d}} = 1$

If

$$(\zeta_n^k)^r = 1, n \mid kr \implies n \mid r \implies \zeta_n^k$$

is a primitive root and there must be $\phi(n)$ of these.

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We can also see pretty clearly from this that

$$\prod \zeta_n^k = 1, n \neq 2$$

using a method that could have easily proven why the totient function is always even: each root could be partitioned into $\{\zeta^k, \zeta^{n-k}$, which clearly multiplies to 1.

Theorem 6.2

The sum of the k^{th} powers of the primitive n^{th} roots of unity is

$$\mu(r)\frac{\phi(n)}{\phi(r)}$$

where $r = \frac{n}{\gcd(n,k)}$ and μ is the Möbius function

Proof.

$$\sum \zeta^k = \mu(k)$$

and there will be $\phi(g)$ of these terms. However, $\phi(n)$ of them are the primitive roots of unity, each being counted $\frac{\phi(g)}{\phi(n)}$ times.

$$\sum \zeta_n^k = \mu(k)\frac{\phi(g)}{\phi(n)}$$

■

7 Resources

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